

Product Price Monitor

Machine Learning Analysis Report

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Executive Summary

This report summarizes automated price monitoring across 169 tracked products over 1 scrape run. Two machine learning techniques were applied to the collected data: anomaly detection, to identify products priced unusually relative to their category, and price forecasting, to project short-term price movement based on historical trend. The analysis identified 42 products with pricing that falls outside the expected range for its category. Only one scrape run available — forecasting requires at least two runs to establish a trend. Results will populate after the next scheduled run.

Anomaly Detection

Each product's latest price is compared against the mean and standard deviation of its own category, using a Z-score threshold of 1.5 standard deviations. To prevent a single extreme value from distorting its own baseline, each product is excluded from the calculation of its category's mean and standard deviation before being scored against it. This avoids a common statistical pitfall where an outlier masks itself by pulling the average toward its own value.

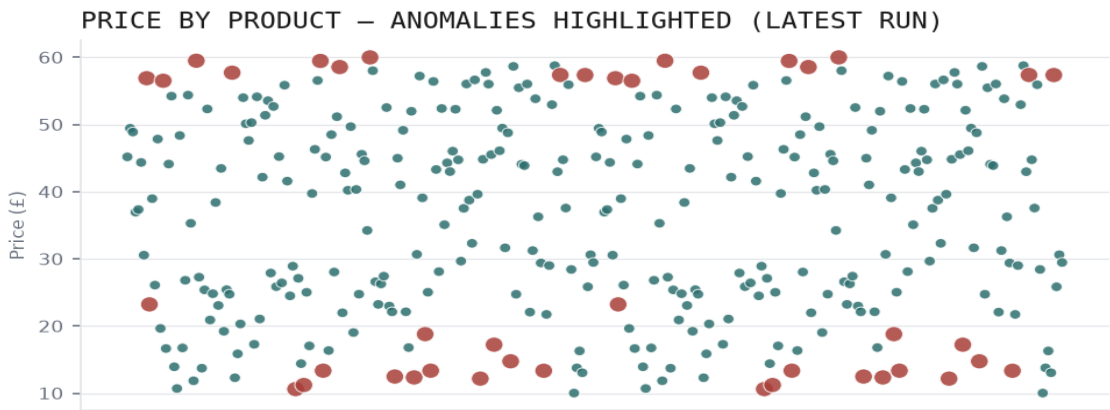


Figure 1. Price by product for the latest scrape run. Each point represents one product; flagged anomalies are shown larger and in a distinct color.

Across the 5 tracked categories, price spread varies considerably: Fiction shows the widest range in this run, from £10.60 to £59.98. The most significant anomaly is “Every Heart a Doorway (Every Heart A Doorway #1)”, priced at £12.16 against a Fantasy category average of £39.59 — a Z-score of -2.00, placing it below the expected range for its category.

Product	Category	Price	Category Avg	Z-Score	Direction
Every Heart a Doorway (Eve...	Fantasy	£12.16	£39.59	-2.00	Below

Every Heart a Doorway (Eve...	Fantasy	£12.16	£39.59	-2.00	Below
Sister Sable (The Mad Quee...	Fantasy	£13.33	£39.59	-1.91	Below
Sister Sable (The Mad Quee...	Fantasy	£13.33	£39.59	-1.91	Below
Princess Between Worlds (W...	Fantasy	£13.34	£39.59	-1.91	Below
Princess Between Worlds (W...	Fantasy	£13.34	£39.59	-1.91	Below
A Year in Provence (Proven...	Travel	£56.88	£39.79	1.86	Above
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Harry Potter and the Chamb...	Fantasy	£14.74	£39.59	-1.81	Below
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The Road to Little Dribbli...	Travel	£23.21	£39.79	-1.80	Below
The Road to Little Dribbli...	Travel	£23.21	£39.79	-1.80	Below
Boar Island (Anna Pigeon #...	Mystery	£59.48	£31.72	1.77	Above
Boar Island (Anna Pigeon #...	Mystery	£59.48	£31.72	1.77	Above
I Am Pilgrim (Pilgrim #1)	Fiction	£10.60	£36.07	-1.73	Below
I Am Pilgrim (Pilgrim #1)	Fiction	£10.60	£36.07	-1.73	Below
Dear Mr. Knightley	Fiction	£11.21	£36.07	-1.69	Below
Dear Mr. Knightley	Fiction	£11.21	£36.07	-1.69	Below
The No. 1 Ladies' Detectiv...	Mystery	£57.70	£31.72	1.65	Above
The No. 1 Ladies' Detectiv...	Mystery	£57.70	£31.72	1.65	Above
Last One Home (New Beginni...	Fiction	£59.98	£36.07	1.62	Above
Last One Home (New Beginni...	Fiction	£59.98	£36.07	1.62	Above
A Feast for Crows (A Song ...	Fantasy	£17.21	£39.59	-1.62	Below
A Feast for Crows (A Song ...	Fantasy	£17.21	£39.59	-1.62	Below
The Silent Wife	Fiction	£12.34	£36.07	-1.61	Below
The Silent Wife	Fiction	£12.34	£36.07	-1.61	Below
Lila (Gilead #3)	Fiction	£12.47	£36.07	-1.60	Below
Lila (Gilead #3)	Fiction	£12.47	£36.07	-1.60	Below
The Improbability of Love	Fiction	£59.45	£36.07	1.58	Above
The Improbability of Love	Fiction	£59.45	£36.07	1.58	Above

Immunity: How Elie Metchni...	Science	£57.36	£33.09	1.57	Above
Immunity: How Elie Metchni...	Science	£57.36	£33.09	1.57	Above
The Disappearing Spoon: An...	Science	£57.35	£33.09	1.57	Above
The Disappearing Spoon: An...	Science	£57.35	£33.09	1.57	Above
The Past Never Ends	Mystery	£56.50	£31.72	1.57	Above
The Past Never Ends	Mystery	£56.50	£31.72	1.57	Above
Mothering Sunday	Fiction	£13.34	£36.07	-1.54	Below
Mothering Sunday	Fiction	£13.34	£36.07	-1.54	Below
Miller's Valley	Fiction	£58.54	£36.07	1.52	Above
Miller's Valley	Fiction	£58.54	£36.07	1.52	Above
Unicorn Tracks	Fantasy	£18.78	£39.59	-1.50	Below
Unicorn Tracks	Fantasy	£18.78	£39.59	-1.50	Below

Table 1. All products flagged as anomalous in the latest run, ranked by absolute Z-score. A positive Z-score indicates a price above the category average; a negative value indicates a price below it.

Price Forecast

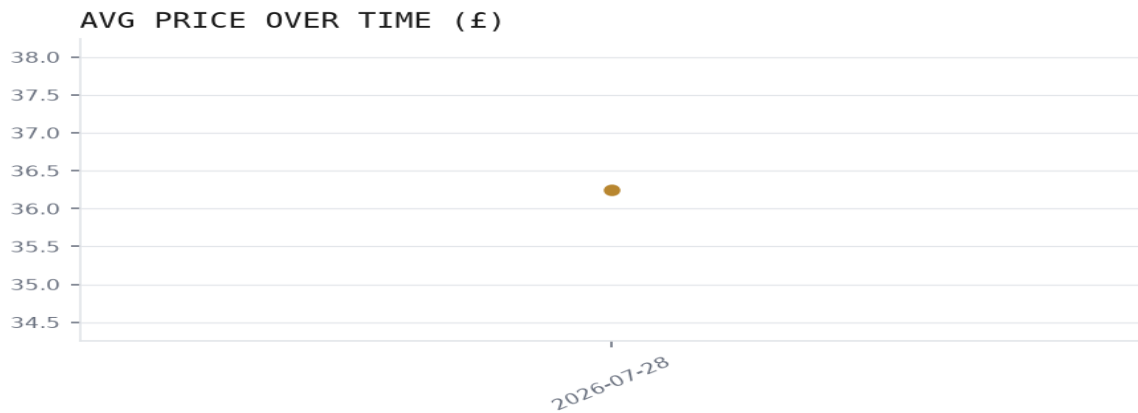
A linear regression model is fit to each product's price history, using scrape run sequence as the independent variable, to project price for the next scheduled run. This is a simple trend-following method: it assumes the rate of change observed so far will continue, and does not account for external market factors. Only one scrape run available — forecasting requires at least two runs to establish a trend. Results will populate after the next scheduled run.

Forecasting requires at least two scrape runs and will populate automatically once additional data has been collected.

Reference Charts

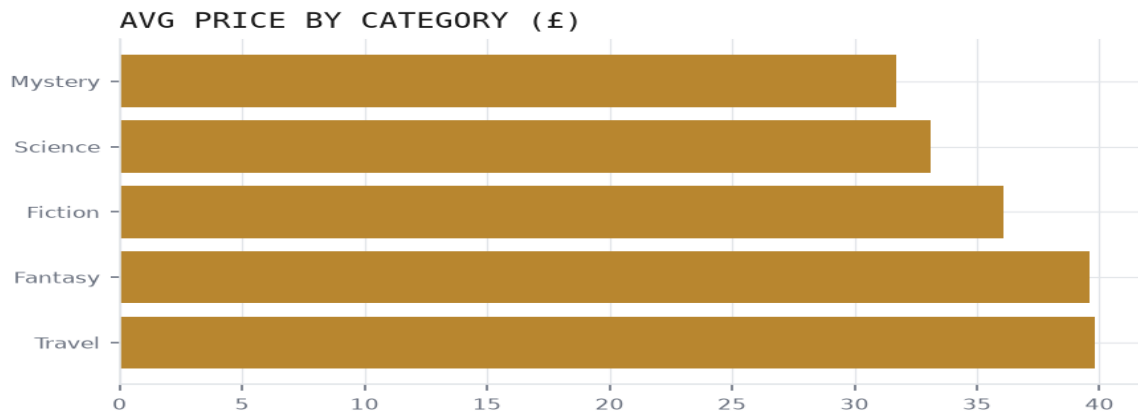
The charts in this section summarize the underlying dataset used in the analysis above and are provided for additional context.

Average Price Over Time



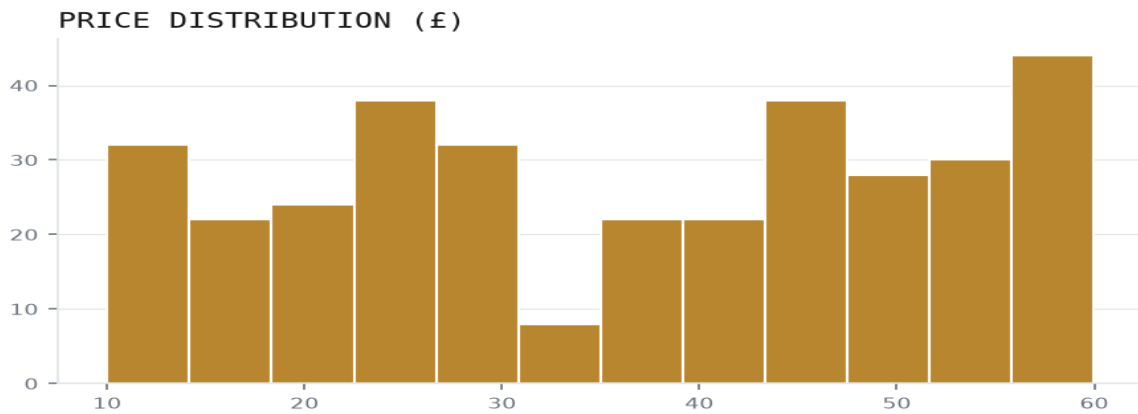
Average price across all tracked products currently stands at £36.24. This is the first recorded run; a trend line will appear once a second run has been collected.

Average Price by Category



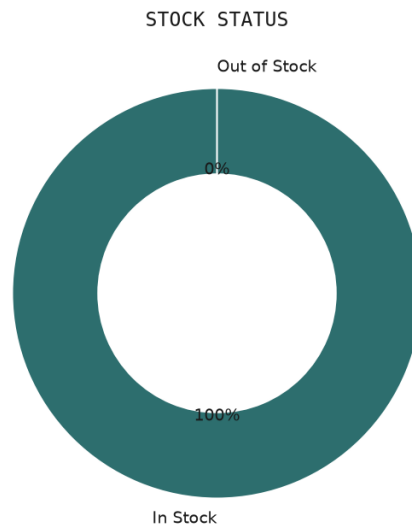
Travel carries the highest average price in this run, at £39.79, while Mystery is the lowest, at £31.72.

Price Distribution



Prices in the latest run range from £10.01 to £59.98, with a median of £37.97. The distribution below shows how tracked products are spread across that range.

Stock Availability



Of the 169 products tracked in the latest run, 340 are currently in stock and 0 are out of stock.